

DistriNVS: Distributional Geometry Transport and Reliability-Guided Completion for Cross-Endoscope Novel View Synthesis

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Abstract—Cross-endoscope novel view synthesis can broaden the visual context available to surgical robots by reconstructing a target view from synchronized RGB-D observations acquired by a spatially displaced endoscope. The task is challenging because partial field of view overlap leaves target regions unobserved, while sparse or erroneous depth makes source support uncertain. In this work, we propose DistriNVS, a framework that separates geometry-supported transport from target-only synthesis. A reliability encoder first estimates source features, depth distributions, confidence, and surface orientation. Distributional surface splatting then propagates these estimates through calibrated geometry and quantifies target view support, depth variance, and collision ambiguity. Guided by this evidence, a reliability-constrained gate combines a bounded transport expert with an uncertainty-conditioned Fourier synthesis expert while preserving supported source content. Extensive experiments on 1392 iMED_NVS test pairs show that DistriNVS achieves the best PSNR, SSIM, and H-PSNR among the compared reconstruction and completion methods. Without target domain adaptation, it also achieves the best PSNR, SSIM, and LPIPS among the compared methods on 275 SCARED test pairs. These results establish reliability-guided transport and synthesis fusion as an effective approach to cross-endoscope novel view synthesis under partial visual overlap. [The code will be made publicly available upon publication.](#)

I. INTRODUCTION

Robot-assisted minimally invasive surgery depends on endoscopic vision for scene perception and instrument guidance [1]. Cross-endoscope novel view synthesis reconstructs a target endoscope image from source endoscope RGB-D observations and the desired source-to-target transformation [2]. However, partial field-of-view overlap means that the target may observe anatomy absent from the source views (Fig. 1). The task therefore combines two distinct regimes: co-visible anatomy is observed in the source view and can be transported through calibrated geometry, whereas target-only regions have no corresponding source observation and require image synthesis. A reliable system should retain source evidence, identify inferred content, and thereby support robotic perception with a wider field of view, retrospective visualization, and surgical simulation [3]. Such uncertainty-aware cross view completion may also support visual state construction for surgical robot [4].

Achieving this separation is difficult because endoscopic depth estimates are unreliable around specularities, weak texture, depth discontinuities, instruments, and deforming tissue [5]. Conventional forward warping represents each valid source pixel by one 3D point; discretized projection leaves sampling gaps, while multiple points may compete for the same target pixel. The binary validity mask conflates missing depth, rasterization holes, depth sorting conflicts, and cross-

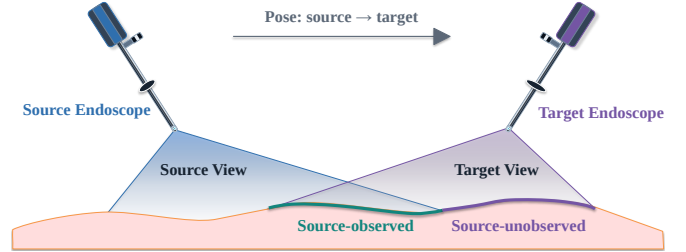


Fig. 1. Cross-endoscope novel view synthesis under partial field-of-view overlap. Given the source to target pose, co-visible anatomy is source-supported and can be transported through calibrated geometry (— teal), whereas target-only anatomy lacks source evidence and needs to be synthesized (— purple).

endoscope disocclusions. Treating these cases uniformly either degrades observed structure or permits hallucination where geometric evidence remains trustworthy.

Existing methods address parts of this problem. Differentiable point rendering and softmax splatting improve feature transport and collision handling [6], while reference-guided inpainting combines depth-based alignment with learned completion [7]. Large mask inpainting networks provide strong global image priors [8], and endoscopic video completion adds depth, temporal, or stereo cues [9]. Endoscopic neural rendering has advanced sparse view reconstruction and uncertainty modeling [10]. However, the approaches address uncertainty before or after view transformation. Deterministic projection assigns the same reliability to all valid depths, while uncertainty in image space characterizes uncertainty in the target image domain. The propagation of depth uncertainty for each source pixel through a spatially displaced cross-endoscope transform remains less explored.

In this paper, we introduce DistriNVS, a reliability-aware framework for direct cross-endoscope novel view synthesis. Its distributional surface splatting (DSS) represents each source depth as a distribution whose bounded mean and scale, surface normal, and confidence are predicted by a depth reliability encoder. DSS projects deterministic depth hypotheses through the calibrated cross-endoscope transform and aggregates source features using footprints with soft nearest surface visibility, yielding target support, depth variance, and collision entropy. These statistics constrain a soft mixture between bounded residual transport for co-visible regions and uncertainty-conditioned Fourier completion for unsupported or ambiguous regions. The paper makes three contributions:

- We propose a reliability-aware framework for cross-endoscope novel view synthesis that separates the trans-

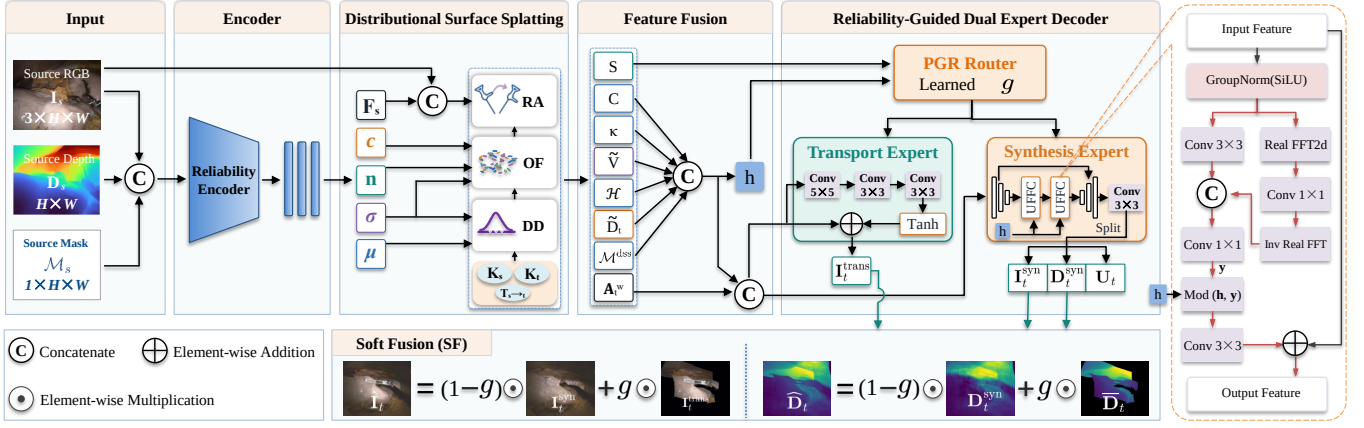


Fig. 2. DistriNVS framework. Encoder outputs $\mathbf{F}_s, c, \mathbf{n}, \sigma, \mu$; DSS outputs $S, C, \kappa, \tilde{V}, \mathcal{H}, \tilde{D}_t, \mathcal{M}^{\text{dss}}, \mathbf{A}_t^w$; and the PGR gate and branch predictions $g, \mathbf{I}_t^{\text{trans}}, \mathbf{I}_t^{\text{syn}}, \mathbf{D}_t^{\text{syn}}, \mathbf{U}_t$ are detailed in Secs. III-A, III-B, and III-C, respectively. The framework outputs fused target RGB $\hat{\mathbf{I}}_t$, fused target depth $\hat{\mathbf{D}}_t$, and target-view risk $\mathbf{U}_t$. In each UFFC block, $\text{Mod}(\mathbf{h}, \mathbf{y})$ modulates the fused local Fourier feature $\mathbf{y}$ using the resized reliability condition $\mathbf{h}$.

port of anatomy observed in the source view from the synthesis of content unobserved in the source view.

- We introduce DSS, a differentiable renderer that propagates depth distributions through calibrated cross-endoscope geometry and exposes target support, depth variance, and collision entropy beyond binary validity.
- We design a decoder with dual experts that forms a soft mixture of bounded residual transport and uncertainty-conditioned Fourier completion under a reliability constraint.

II. RELATED WORK

We summarize image completion, geometry-guided view transport, and endoscopic neural rendering, with particular attention to how existing approaches represent missing content, exploit calibrated geometry, and quantify uncertainty.

A. Image and Endoscopic Video Inpainting

Generic image inpainting combines mask-aware feature extraction with broad appearance priors. Partial and gated convolutions prevent invalid activations from contaminating local features [11], whereas LaMa and UFFC use Fourier operations for global context and MAT restricts attention to valid tokens [8], [12], [13]. Recent diffusion models such as BrushNet and TurboFill incorporate masked image features into pretrained generators or adapt sampling with few steps for efficient completion [14], [15]. Despite strong appearance priors, these methods define holes in a fixed image coordinate system and do not describe geometric provenance. Endoscopic completion adds domain cues: DAEVI combines spatiotemporal RGB and depth, and SSIFNet fuses flow and stereo temporal information for instrument occlusions [9], [16]. Such methods exploit neighboring frames or views but still restore defects defined in the original camera coordinates. Cross-endoscope synthesis instead creates holes through viewpoint transfer, while rasterization gaps or depth conflicts may retain source support. In this paper, the proposed DistriNVS uses calibrated projection statistics to

distinguish true disocclusions from rasterization gaps and depth conflicts.

B. Geometry-Guided Transport and Completion

Geometry-guided view synthesis preserves correspondence by transporting source evidence before generating missing content. SynSin predicts and projects a feature point cloud, softmax splatting resolves many-to-one mappings with learned importance, and GeoFill refines depth and pose before reprojection and inpainting [6], [7], [17]. Recent feedforward Gaussian renderers use stronger explicit representations. MVSplat localizes Gaussians with plane-sweep cost volumes, whereas DepthSplat combines multi-view correspondence with monocular depth priors [18], [19]. They reduce the need for optimization for each scene but infer a radiance representation from multiple calibrated observations, without an explicit per-pixel boundary between measured and synthesized appearance. Generative methods address wider view gaps by synthesizing missing content from reference or geometric cues. LeftRefill frames synthesis conditioned on a reference as paired canvas completion, MultiDiff conditions video diffusion on warped depth reference views, and latent diffusion guided by points uses clouds derived from stereo for laparoscopic view synthesis [20]–[22]. These models can synthesize missing regions but do not distinguish whether a target pixel is supported by reliable source geometry. In contrast, DistriNVS estimates support from the source view for each target pixel, preserves transported source content in reliable regions, and applies synthesis to unobserved regions.

C. Endoscopic Novel View Synthesis and Uncertainty

Endoscopic neural rendering handles nonrigid tissue, illumination, tool occlusions, and sparse camera coverage. EndoNeRF represents deformation with a neural radiance field, while EndoSurf combines dynamic modeling with a neural surface [23], [24]. EndoSparse introduces geometric and appearance priors into Gaussian splatting, and EndoGSMT uses depth, dense displacement, and motion priors

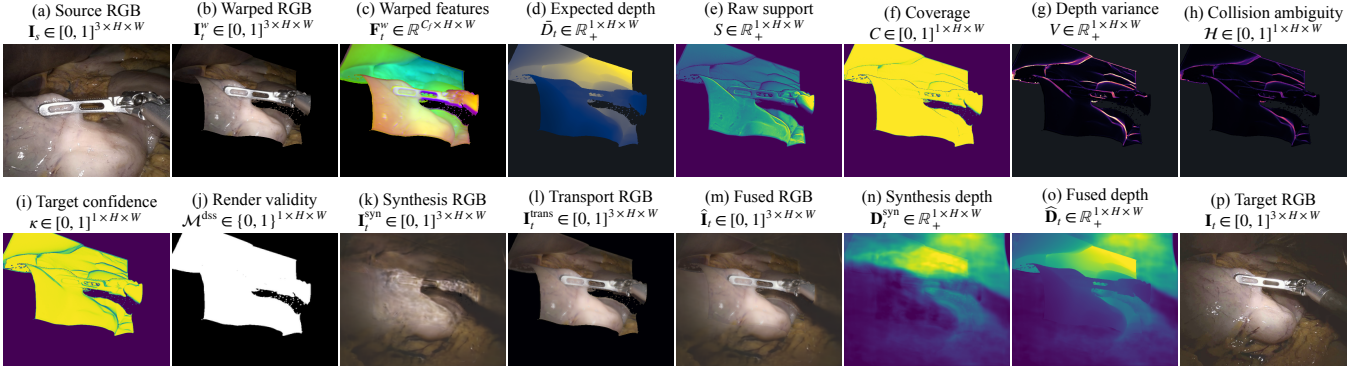


Fig. 3. DSS evidence and network predictions for a representative iMED-NVS pair. (a) Source RGB. (b)–(j) Transported attributes and reliability statistics produced by DSS. (k)–(o) Synthesis, transport, and fused RGB-D predictions. (p) Target RGB. Scalar maps are pseudocolored for visualization.

for dynamic reconstruction [25], [26]. GastroNVS further shows that large viewpoint changes in real gastroscopy amplify geometric artifacts and illumination mismatch between views [27]. These systems build persistent scene representations from multiple frames and address reconstruction fidelity within a captured scene. There also exist uncertainty-aware approaches for addressing sparse evidence in endoscopic scenes. UC-NeRF conditions a radiance field on multi-view uncertainty, whereas ExtraGS uses uncertainty to sample virtual cameras and weight pseudo observations refined by diffusion for extrapolation [3], [10]. In contrast, we propagate a distribution for each input depth through the calibrated cross-endoscope transform and derive target support, depth variance, and collision entropy from the same rasterization that transports appearance.

III. METHOD

DistriNVS receives source RGB $\mathbf{I}_s \in \mathbb{R}^{1 \times H \times W}$, metric depth $\mathbf{D}_s \in \mathbb{R}_+^{1 \times H \times W}$, validity mask $\mathcal{M}_s \in \{0, 1\}^{1 \times H \times W}$, source and target intrinsics $\mathbf{K}_s, \mathbf{K}_t \in \mathbb{R}^{3 \times 3}$, and the rigid cross-endoscope transform $\mathbf{T}_{s \rightarrow t} \in \text{SE}(3) \subset \mathbb{R}^{4 \times 4}$. The model predicts target RGB $\hat{\mathbf{I}}_t$, depth $\hat{\mathbf{D}}_t$, and risk $\mathbf{U}_t \in [0, 1]^{1 \times H \times W}$. A reliability encoder first estimates source features, geometry, and uncertainty. DSS transports these attributes into the target view and derives expected depth, support, and ambiguity. The transported attributes and reliability maps condition the transport expert, synthesis expert, and router. The router produces a reliability-constrained soft gate that fuses the two expert predictions into the final target RGB-D. The synthesis head estimates target-view risk. We denote the reliability encoder, depth-distribution construction, orientation-aware footprint computation, reliability aggregation, and expert prediction by f_{enc} , f_{DD} , f_{OF} , f_{RA} , and f_{exp} , respectively. Fig. 2 summarizes the pipeline of the proposed method.

A. Reliability Encoder

The reliability encoder constructs the source geometric representation for DSS by modeling the reliability of measured and inferred depths. A branch at full resolution preserves local boundaries through residual convolutional blocks, while a branch at quarter resolution captures broader

context using strided convolution. The context features are upsampled using bilinear interpolation, concatenated with the features from the branch at full resolution, and fused through a residual connection. A shared prediction head produces a feature map $\mathbf{F}_s \in \mathbb{R}^{C_f \times H \times W}$ and raw depth, scale, confidence, and orientation parameters. These parameters yield depth fields $\mu, \sigma \in \mathbb{R}_+^{1 \times H \times W}$, confidence $c \in [0, 1]^{1 \times H \times W}$, and a unit-length orientation proxy $\mathbf{n} \in \mathbb{R}^{3 \times H \times W}$. We denote the complete encoder mapping by

$$(\mathbf{F}_s, \mu, \sigma, c, \mathbf{n}) = f_{\text{enc}}(\mathbf{I}_s, \mathbf{D}_s, \mathcal{M}_s). \quad (1)$$

The shared prediction head applies bounded residual corrections to measured depths and infers missing depths through a *depth completion* (DC) parameterization. When DC is enabled, both measured and completed depths participate in DSS, with reduced confidence assigned to completed estimates. The orientation proxy is learned under the downstream objectives and is used to adapt the splat footprint. During training, random depth dropout exposes the encoder to both measured and missing depth patterns, while depth supervision is retained on the originally valid samples.

B. Distributional Surface Splatting

DSS comprises Distributional Depth (DD), Orientation-aware Footprints (OF), and Reliability Aggregation (RA). DD propagates source depth uncertainty through the calibrated cross-endoscope transform using deterministic depth hypotheses. OF adapts each projected footprint to depth uncertainty and surface orientation. RA aggregates the transported hypotheses and derives reliability statistics in the target view for downstream routing.

Distributional depth (DD). Given the predicted mean μ and scale σ , DD constructs three depth hypotheses using fixed offsets $\xi_j \in \{-1, 0, 1\}$ and normalized Gaussian weights a_j , where $\sum_{j=1}^3 a_j = 1$. The calibrated pinhole operator $\Pi_{s \rightarrow t}$ back-projects each hypothesis with $\mathbf{K}_s$, applies $\mathbf{T}_{s \rightarrow t}$, and projects it with $\mathbf{K}_t$. Specifically, $z_j = \mu + \xi_j \sigma$ and $(\mathbf{X}_j, D_j^t) = \Pi_{s \rightarrow t}(z_j, \mathbf{K}_s, \mathbf{K}_t, \mathbf{T}_{s \rightarrow t})$ for $j \in \{1, 2, 3\}$. The complete DD mapping is

$$\{z_j, a_j, \mathbf{X}_j, D_j^t\}_{j=1}^3 = f_{\text{DD}}(\mu, \sigma, \mathbf{K}_s, \mathbf{K}_t, \mathbf{T}_{s \rightarrow t}), \quad (2)$$

where $z_j, D_j^t \in \mathbb{R}_+^{1 \times H \times W}$ denote the source depth hypothesis and its depth in the target frame, respectively, and

$\mathbf{X}_j \in \mathbb{R}^{2 \times H \times W}$ contains the projected target coordinates [6], [17].

Orientation-aware footprints (OF). For each projected hypothesis, OF constructs an elliptical footprint based on projection sensitivity, predicted depth uncertainty, and surface orientation, with $\mathbf{r}_j = b + |\mathbf{J}_j| \odot \sigma + b_n(1 - |n^z|)$. The complete OF mapping is

$$\{\mathbf{r}_j\}_{j=1}^3 = f_{\text{OF}}(\{\mathbf{X}_j, D_j^t\}_{j=1}^3, \sigma, \mathbf{n}; b, b_n), \quad (3)$$

where $\mathbf{J}_j, \mathbf{r}_j \in \mathbb{R}^{2 \times H \times W}$ denote the projection sensitivity along two axes and footprint radii, respectively, while b and b_n control the base expansion and the expansion based on orientation. Each hypothesis is splatted using weights determined by confidence c , the hypothesis weight a_j , distance within the elliptical footprint, and soft nearest-surface depth ordering. Nearest-depth selection and integer assignment are discrete; the remaining operations are differentiable.

Reliability aggregation (RA). RA uses the distributional footprints to transport the concatenated source attributes $\mathbf{A}_s = [\mathbf{I}_s, \mathbf{F}_s] \in \mathbb{R}^{(3+C_f) \times H \times W}$. RA aggregates overlapping hypotheses to produce $\mathbf{A}_t^w = [\mathbf{I}_t^w, \mathbf{F}_t^w] \in \mathbb{R}^{(3+C_f) \times H \times W}$, the expected depth $\bar{D}_t$, and the conditional depth variance V . Raw support is mapped to bounded coverage C and render validity $\mathcal{M}^{\text{dss}}$. Collision ambiguity $\mathcal{H}$ is computed from normalized weight entropy and depth dispersion, discounting benign subpixel overlap. Target confidence κ is derived from $(C, V, \mathcal{H})$. The complete aggregation is

$$\begin{aligned} &(\mathbf{A}_t^w, \bar{D}_t, S, V, C, \mathcal{H}, \kappa, \mathcal{M}^{\text{dss}}) \\ &= f_{\text{RA}}(\mathbf{A}_s, c, \{\mathbf{X}_j, D_j^t, a_j, \mathbf{r}_j\}_{j=1}^3; \tau_z), \end{aligned} \quad (4)$$

where S denotes raw geometric support, V depth variance, C bounded coverage, $\mathcal{H}$ collision ambiguity, and κ target confidence. Together, these statistics provide continuous geometric evidence beyond the binary render validity mask. Fig. 3 visualizes these DSS outputs alongside the downstream network predictions.

C. Reliability-Guided Dual Expert Fusion

Physics-guided routing (PGR). PGR forms the reliability condition $\mathbf{h} = [C, \kappa, \tilde{V}, \mathcal{H}, \tilde{D}_t, \mathcal{M}^{\text{dss}}] \in \mathbb{R}^{6 \times H \times W}$, where $\tilde{V}$ is the normalized form of V and $\tilde{D}_t$ is the normalized form of $\bar{D}_t$. A synthesis prior $g^0 \in [0, 1]^{1 \times H \times W}$ derived from physics increases with decreasing confidence, increasing depth dispersion, and increasing collision ambiguity. For render-valid pixels, the prior is computed from the weighted reliability terms, whereas pixels without DSS support are assigned a synthesis prior of one, yielding $g^0 = (1 - \mathcal{M}^{\text{dss}}) + \mathcal{M}^{\text{dss}} \odot [\lambda_\kappa(1 - \kappa) + \lambda_V \tilde{V} + \lambda_H \mathcal{H}]$. Before conversion to logit space, the prior is clipped to $[\epsilon, 1 - \epsilon]$ for numerical stability. The reliable support mask $\mathcal{M}^{\text{rel}} \in \{0, 1\}^{1 \times H \times W}$, obtained as $\mathcal{M}^{\text{rel}} \leftarrow (S, V, \mathcal{H}, \mathcal{M}^{\text{dss}})$, marks DSS evidence as reliable when the render is valid, support is sufficient, and depth variance and collision ambiguity remain below their thresholds. The router adds a bounded residual $\Delta_\theta(\mathbf{h}) \in \mathbb{R}^{1 \times H \times W}$ to the detached prior logit, where $\text{sg}(\cdot)$ denotes

the stop-gradient operation. The effective synthesis gate $g \in [0, 1]^{1 \times H \times W}$ is

$$g = (1 - \mathcal{M}^{\text{rel}}) \odot \text{sigmoid}(\text{sg}[\text{logit}(g^0)] + \Delta_\theta(\mathbf{h})). \quad (5)$$

Complementary experts. We employ a *transport expert* to correct source-supported appearance and a *synthesis expert* to generate content where geometric evidence is unreliable. We form the shared expert input by concatenating the transported content and reliability evidence as $\mathbf{X}_t = [\mathbf{A}_t^w, \mathbf{h}] \in \mathbb{R}^{(C_f+9) \times H \times W}$. Given $(\mathbf{X}_t, \mathbf{I}_t^w)$, the transport expert predicts a bounded RGB correction to $\mathbf{I}_t^w$ while retaining the DSS expected depth $\bar{D}_t$. The synthesis expert processes $(\mathbf{X}_t, \mathbf{h})$ with a Fourier U-Net operating at four resolutions and conditioned on uncertainty [28]. A 5×5 stem first projects $\mathbf{X}_t$ to 32 channels. The resulting features are then processed by three stride-2 encoder stages and three symmetric decoder stages with bilinear upsampling and skip connections. Each stage contains an uncertainty-conditioned Fourier convolution (UFFC) block. The block sends normalized features through a local 3×3 convolution branch and a parallel global Fourier branch. The Fourier branch applies a 2D real FFT [8] and mixes the concatenated real and imaginary coefficients using a 1×1 convolution. An inverse real FFT maps the mixed coefficients back to the spatial domain. A 1×1 convolution fuses the outputs of the two branches. At every resolution, a learned affine transform derived from the resized condition tensor $\mathbf{h}$ modulates the fused features before residual addition. A final 3×3 head jointly predicts RGB logits, normalized depth, and risk. The expert outputs are $\mathbf{I}_t^{\text{trans}}$, $\mathbf{I}_t^{\text{syn}}$, $\mathbf{D}_t^{\text{syn}}$, and $\mathbf{U}_t \in [0, 1]^{1 \times H \times W}$. We denote the joint expert mapping by

$$(\mathbf{I}_t^{\text{trans}}, \mathbf{I}_t^{\text{syn}}, \mathbf{D}_t^{\text{syn}}, \mathbf{U}_t) = f_{\text{exp}}(\mathbf{X}_t, \mathbf{I}_t^w, \mathbf{h}). \quad (6)$$

Reliability-constrained output. The transport branch retains source evidence while allowing bounded appearance correction; the synthesis branch supplies content where geometric evidence is weak. The fused RGB prediction is

$$\hat{\mathbf{I}}_t = (1 - g) \odot \mathbf{I}_t^{\text{trans}} + g \odot \mathbf{I}_t^{\text{syn}}. \quad (7)$$

The fused depth prediction is

$$\hat{\mathbf{D}}_t = (1 - g) \odot \bar{D}_t + g \odot \mathbf{D}_t^{\text{syn}}. \quad (8)$$

The gate is broadcast across RGB channels. We refer to this output rule as *soft fusion (SF)*. Within reliable support regions indicated by $\mathcal{M}^{\text{rel}}$, $g = 0$, and the output is determined by the bounded transport expert; elsewhere the two experts form a soft mixture. For comparison, *hard composition (HC)* overrides the SF result in the reliable support regions indicated by $\mathcal{M}^{\text{rel}}$ by copying the unmodified DSS RGB $\mathbf{I}_t^w$ and depth $\bar{D}_t$, while retaining the SF result elsewhere.

D. Learning Objectives

Regional reconstruction. During training, target RGB $\mathbf{I}_t$ and target depth $\mathbf{D}_t \in \mathbb{R}_+^{1 \times H \times W}$ provide supervision. A tissue-valid mask $\mathcal{M}_t^{\text{tis}} \in \{0, 1\}^{1 \times H \times W}$ excludes annotated instruments when available. We define the predicted

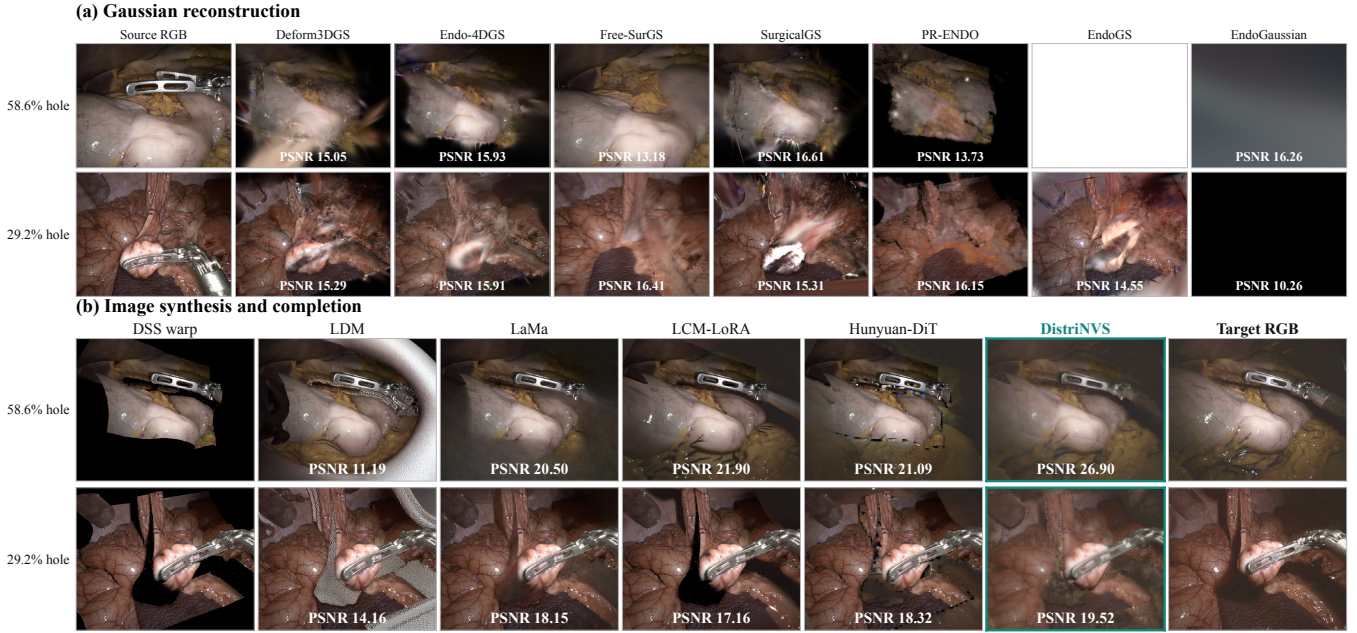


Fig. 4. Qualitative in-domain comparison on iMED_NVS. (a) Gaussian reconstruction from source RGB. (b) Image completion from the DSS warp. White labels report full image PSNR (dB). In the second scene, residual tool misalignment may arise from errors in the camera pose provided by the dataset.

TABLE I

FULL IMAGE QUANTITATIVE COMPARISON ON THE iMED_NVS TEST SET ($N = 1,392$).

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Deform3DGS [29]	15.83 ± 1.87	0.33 ± 0.06	0.48 ± 0.10
Endo-4DGS [30]	16.78 ± 1.65	0.36 ± 0.04	0.46 ± 0.08
Free-SurGS [31]	17.09 ± 2.14	0.43 ± 0.07	0.51 ± 0.07
SurgicalGS [32]	17.79 ± 1.95	0.38 ± 0.07	0.46 ± 0.07
PR-ENDO [33]	15.79 ± 1.27	0.31 ± 0.07	0.55 ± 0.07
EndoGS [34]	12.52 ± 6.22	0.38 ± 0.06	0.66 ± 0.20
EndoGaussian [35]	14.38 ± 4.26	0.37 ± 0.24	0.77 ± 0.12
LDM [36]	17.18 ± 2.83	0.34 ± 0.06	0.44 ± 0.08
LaMa [8]	19.48 ± 1.44	0.43 ± 0.11	0.35 ± 0.06
LCM-LoRA [37]	19.47 ± 1.73	0.42 ± 0.12	0.39 ± 0.04
Hunyuan-DiT [38]	<u>19.52 ± 1.51</u>	0.42 ± 0.12	<u>0.38 ± 0.04</u>
DistriNVS (SF)	22.95 ± 2.33	0.59 ± 0.11	0.40 ± 0.05

TABLE II

UNOBSERVED REGION QUANTITATIVE COMPARISON ON THE iMED_NVS TEST SET ($N = 1,392$).

Method	H-PSNR $\uparrow$	H-SSIM $\uparrow$	H-LPIPS $\downarrow$
Deform3DGS [29]	14.01 ± 1.81	0.31 ± 0.07	0.63 ± 0.06
Endo-4DGS [30]	15.79 ± 1.85	0.42 ± 0.14	0.60 ± 0.06
Free-SurGS [31]	17.86 ± 2.88	0.56 ± 0.05	0.56 ± 0.06
SurgicalGS [32]	17.63 ± 2.87	0.47 ± 0.16	0.59 ± 0.06
PR-ENDO [33]	11.92 ± 1.71	0.10 ± 0.06	0.74 ± 0.08
EndoGS [34]	12.52 ± 6.68	0.49 ± 0.12	0.74 ± 0.17
EndoGaussian [35]	15.86 ± 3.60	0.48 ± 0.31	0.69 ± 0.09
LDM [36]	14.90 ± 3.48	0.37 ± 0.11	0.64 ± 0.13
LaMa [8]	21.40 ± 2.32	0.63 ± 0.05	0.42 ± 0.08
LCM-LoRA [37]	22.53 ± 2.23	0.70 ± 0.06	0.42 ± 0.04
Hunyuan-DiT [38]	20.87 ± 2.07	0.64 ± 0.08	0.49 ± 0.03
DistriNVS (SF)	27.95 ± 2.91	0.80 ± 0.07	0.41 ± 0.05

unreliable-region mask as $\mathcal{M}^{\text{hole}} = 1 - \mathcal{M}^{\text{rel}}$, where $\mathcal{M}^{\text{hole}} \in \{0, 1\}^{1 \times H \times W}$. The unreliable, trusted, and transition tissue masks are defined as $\mathcal{M}_u = \mathcal{M}^{\text{hole}} \cap \mathcal{M}_t^{\text{tis}}$, $\mathcal{M}_v = \mathcal{M}^{\text{rel}} \cap \mathcal{M}_t^{\text{tis}}$, and $\mathcal{M}_b = \text{Band}(\mathcal{M}^{\text{hole}}) \cap \mathcal{M}_t^{\text{tis}}$, respectively; all three lie in $\{0, 1\}^{1 \times H \times W}$. The regional reconstruction loss $\mathcal{L}_{\text{rec}}(\hat{\mathbf{I}}_t, \mathbf{I}_t; \mathcal{M}_u, \mathcal{M}_v, \mathcal{M}_b)$ applies robust Charbonnier RGB penalties to these regions and a gradient penalty on $\mathcal{M}_u \cup \mathcal{M}_b$ [39].

Depth supervision. The depth loss $\mathcal{L}_{\text{depth}}$ is defined over $(\mu, \sigma, \hat{\mathbf{D}}_t; \mathbf{D}_s^{\text{gt}}, \mathcal{M}_s^{\text{gt}}, \mathbf{D}_t, \mathcal{M}_t^D)$, where $\mathbf{D}_s^{\text{gt}} \in \mathbb{R}_+^{1 \times H \times W}$ and $\mathcal{M}_s^{\text{gt}}, \mathcal{M}_t^D \in \{0, 1\}^{1 \times H \times W}$. It combines a Laplace negative log-likelihood over the originally valid source samples with a relative target depth error evaluated over $\mathcal{M}_t^D$ when target depth is available [40].

Cycle consistency and routing. For *RGB-D-guided re-projection cycle consistency (RCC)*, the predicted RGB-D is splatted through the inverse calibration to obtain backward-

rendered RGB $\mathbf{I}_s^{\text{back}} \in [0, 1]^{3 \times H \times W}$. The cycle loss is

$$\mathcal{L}_{\text{RCC}} = \langle \rho_{\epsilon}(\mathbf{I}_s^{\text{back}} - \mathbf{I}_s) \rangle_{\mathcal{M}_s^{\text{cyc}}}, \quad (9)$$

where $\rho_{\epsilon}(x) = \sqrt{x^2 + \epsilon^2}$ is the Charbonnier penalty, $\epsilon = 10^{-3}$, and $\langle \cdot \rangle_{\mathcal{M}_s^{\text{cyc}}}$ denotes averaging over valid cycle pixels identified by $\mathcal{M}_s^{\text{cyc}} \in \{0, 1\}^{1 \times H \times W}$ [39]. Risk modulates the backward splat, but no depth cycle loss is imposed. The routing loss uses binary cross-entropy between the pre-sigmoid synthesis logits and a detached binary target derived from $(S, V, \mathcal{H})$.

Uncertainty supervision and total objective. The *uncertainty loss (UL)* trains the predicted risk as an error scale within the unreliable regions defined by $\mathcal{M}_u$. Let $\mathbf{E}$ be the channel-mean absolute RGB error map and $\mathbf{s}$ the positive scale map obtained from $\mathbf{U}_t$ through a bounded affine transform. We use $\mathcal{L}_{\text{UL}} = \langle \mathbf{E}/\mathbf{s} + \log \mathbf{s} \rangle_{\mathcal{M}_u}$. The

complete objective groups these terms as

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_d \mathcal{L}_{\text{depth}} + \lambda_c \mathcal{L}_{\text{RCC}} + \lambda_r \mathcal{L}_{\text{router}} + \lambda_u \mathcal{L}_{\text{UL}}, \quad (10)$$

where $\lambda_d, \lambda_c, \lambda_r, \lambda_u \in \mathbb{R}_+$ are weights that balance the depth, cycle-consistency, routing, and uncertainty terms, respectively.

IV. EXPERIMENTS

A. Experimental Setup

Datasets. iMED_NVS contains 20 scenes and is split at the scene level into 13 training scenes with 2561 synchronized pairs and 7 test scenes with 1392 pairs [41]. Datasets 8 and 9 of SCARED [42] are used for zero-shot evaluation. Each synchronized pair comprises a right camera RGB-D source and a left camera RGB target; DSS uses their intrinsics and calibrated relative camera pose. We partition each sequence into temporal windows of 10 frames and select the frame nearest to each window center while maintaining a separation of at least five frames. This protocol yields 275 synchronized test pairs.

Comparison Methods. We compare against seven endoscopic Gaussian reconstruction methods [29]–[35] and four feedforward completion systems: LDM, LaMa, LCM-LoRA, and Hunyuan-DiT [8], [36]–[38]. Every prediction is evaluated at the same resolution on the same target frames.

Evaluation Metrics. We report PSNR [43], SSIM [44], and LPIPS [45]. H-prefixed metrics are evaluated on the canonical raw DSS hole intersected with non-tool tissue support on iMED_NVS. All SCARED pixels are evaluated because target tool masks are unavailable. Latency is measured at batch size one on an RTX 4090. In evaluation, degenerate predictions are retained, and all predictions are evaluated **without** postprocessing.

Implementation. Images are processed at spatial resolution $H \times W = 512 \times 640$. DSS uses the three Gaussian-weighted hypotheses described in Sec. III, with $(a_j) = (0.2741, 0.4518, 0.2741)$, $b = b_n = 0.75$ pixels, and $\tau_z = 2$ mm. The routing prior weights are $(\lambda_K, \lambda_V, \lambda_H) = (0.65, 0.20, 0.15)$, with $\epsilon = 10^{-4}$ for clipping g^0 before the logit transform. The reliability encoder transports 16 feature channels. The synthesis expert has four resolutions and base width 32, and the complete network contains 5.79M trainable parameters. We optimize for 10^4 epochs using AdamW with a learning rate of 2×10^{-4} , weight decay 10^{-4} , and effective batch size 12. The loss weights are $(\lambda_d, \lambda_c, \lambda_r, \lambda_u) = (0.2, 0.2, 0.1, 0.1)$. Training is performed on 4 NVIDIA RTX 4090 GPUs using BF16 mixed precision and seed 6666. Renderer, routing, mask, and loss constants are fixed across corresponding configurations.

B. In-Domain Comparison

Tab. I and II compare all methods on the same 1392 iMED_NVS frames using shared tissue and hole masks. The Gaussian systems use dataset depth and serve as reconstruction references, whereas DistriNVS is evaluated by its direct SF output. On the iMED_NVS test set, DistriNVS achieved the highest PSNR, SSIM, and H-PSNR while remaining

TABLE III

ZERO-SHOT COMPARISON ON SCARED DATASETS 8 AND 9 ($N = 275$).

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	H-PSNR $\uparrow$	Latency (s) $\downarrow$
LDM	17.44 $\pm$ 1.86	0.33 $\pm$ 0.14	0.81 $\pm$ 0.17	12.54 $\pm$ 2.44	2.57 $\pm$ 0.15
LaMa	22.72 $\pm$ 2.10	0.63 $\pm$ 0.14	0.32 $\pm$ 0.07	21.43$\pm$2.71	0.04$\pm$0.00
LCM-LoRA	14.26 $\pm$ 0.94	0.22 $\pm$ 0.04	1.00 $\pm$ 0.08	14.23 $\pm$ 2.61	0.40 $\pm$ 0.04
Hunyuan-DiT	15.79 $\pm$ 2.43	0.31 $\pm$ 0.12	0.85 $\pm$ 0.15	13.24 $\pm$ 3.21	13.04 $\pm$ 0.07
DistriNVS	23.47$\pm$2.01	0.66$\pm$0.12	0.23$\pm$0.04	21.11$\pm$2.93	0.08$\pm$0.01

TABLE IV

ABLATION STUDY ON THE iMED_NVS TEST SET ($N = 1,392$).

Variant	PSNR $\uparrow$	SSIM $\uparrow$	H-PSNR $\uparrow$	H-SSIM $\uparrow$
Primary output rule and legacy reference				
DistriNVS (SF)	22.95 $\pm$ 2.33	0.59 $\pm$ 0.11	27.95 $\pm$ 2.91	0.80 $\pm$ 0.07
DistriNVS (HC)	<u>20.77 $\pm$ 1.91</u>	<u>0.49 $\pm$ 0.13</u>	27.56 $\pm$ 2.73	0.78 $\pm$ 0.07
Component variants under legacy HC				
w/o DC	20.68 $\pm$ 1.89	0.49 $\pm$ 0.13	27.76 $\pm$ 2.48	0.79 $\pm$ 0.06
w/o DD	20.67 $\pm$ 1.91	0.48 $\pm$ 0.13	27.19 $\pm$ 2.74	0.77 $\pm$ 0.07
w/o OF	20.65 $\pm$ 1.92	0.48 $\pm$ 0.13	27.73 $\pm$ 2.71	0.78 $\pm$ 0.06
w/o PGR	20.75 $\pm$ 1.89	0.49 $\pm$ 0.13	27.41 $\pm$ 2.72	0.77 $\pm$ 0.07
w/o RCC	20.72 $\pm$ 1.91	0.49 $\pm$ 0.13	27.41 $\pm$ 2.73	0.78 $\pm$ 0.07
w/o UL	20.70 $\pm$ 1.89	0.48 $\pm$ 0.13	27.02 $\pm$ 2.60	0.77 $\pm$ 0.07

competitive in LPIPS. Fig. 4 presents the corresponding qualitative evidence. Its examples are the 100th retained pair from each of the lexicographically first and last test scenes, a rule fixed before prediction inspection. We observe in Fig. 4 that the diffusion-based LCM-LoRA achieves the most accurate reconstruction in the shadowed regions. Fig. 6 stratifies reconstruction quality by the fraction of target tissue unobserved from the source view, termed the *unobserved ratio*. DistriNVS achieves the highest PSNR across all unobserved ratio ranges and the lowest LPIPS in the largest missing regime, indicating greater robustness under larger displacements between views and reduced overlap.

C. Zero-Shot Comparison

We compare the five feedforward methods on the SCARED dataset, without SCARED adaptation. Bold and underlining in Tab. III mark the best and second-best unrounded means. On SCARED, DistriNVS achieved the highest PSNR and SSIM, the lowest LPIPS, and the second-highest H-PSNR. It also had the second lowest latency, while LaMa achieved the highest H-PSNR and the lowest latency. The slower inference of DistriNVS is partly due to the computational cost of DSS, whose distributional projection and aggregation operations add overhead beyond the synthesis network itself. Fig. 5 shows the qualitative results of the zero-shot comparison on the SCARED datasets.

D. Ablation Study

Tab. IV presents ablation results on the iMED_NVS test set. The primary model uses SF, whereas the six component variants retain legacy HC. SF achieved the strongest overall reconstruction in Tab. IV. Within the HC mode, removing DD causes the clearest joint degradation, whereas DC, OF, PGR, RCC, and UL show smaller or region-dependent changes.

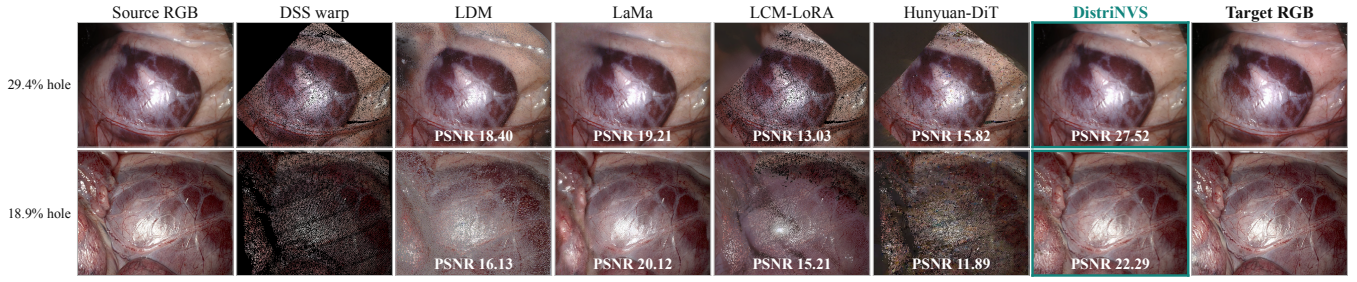


Fig. 5. Qualitative zero-shot comparison on SCARED dataset without adaptation to the target domain. All completion baselines use the same DSS warp and unsupported region mask. White labels report full image PSNR (dB). Hole percentages denote the fraction of target pixels without raw DSS support.

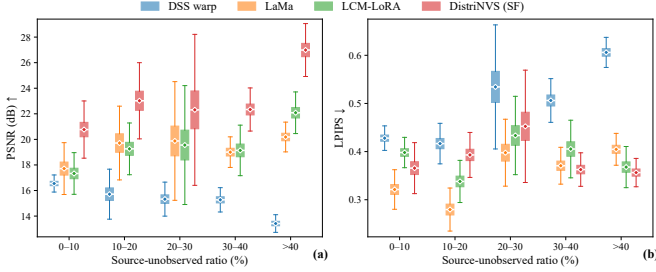


Fig. 6. Performance stratified by the unobserved ratio on the iMED_NVS test set. Curves and shaded regions show the frame-wise mean and sample standard deviation, respectively, for (a) PSNR and (b) LPIPS.

E. Performance Analysis

We observe a remaining gap to LaMa on several specific metrics, which may stem from differences in model capacity and training objectives. LaMa uses approximately 51M parameters and combines global Fourier processing with adversarial, feature matching, and perceptual objectives tailored to image inpainting. By contrast, our model uses only 5.79M parameters while allocating capacity to geometry transport, reliability estimation, routing, RGB-D prediction, and uncertainty modeling. Endoscopic tissue often exhibits smooth color variation and texture dominated by low frequencies, which may favor LaMa’s larger inpainting network. This difference may explain its lower LPIPS on iMED_NVS and its H-PSNR advantage on SCARED. Our results nevertheless show that DistriNVS achieves better zero-shot PSNR, SSIM, and LPIPS on SCARED while using about one ninth as many parameters.

V. CONCLUSION

In this paper, we introduce DistriNVS, a reliability-aware framework for cross-endoscope novel view synthesis that separates geometry-supported transport from target-only synthesis. DistriNVS combines distributional surface splatting, which propagates depth uncertainty and estimates target view reliability, with reliability-constrained expert fusion that preserves supported evidence while completing unobserved regions. Extensive experiments on iMED_NVS and zero-shot SCARED show that DistriNVS outperforms Gaussian reconstruction and image completion baselines, particularly when large target regions are unobserved by the source view.

REFERENCES

[1] R. Wei, K. Chen, Y. L. Ng, Y. Ma, J. D.-L. Ho, H. S. Tong, X. Wang, J. Dai, K.-W. Kwok, and Q. Dou, “Real-time monocular 2-d and 3-d perception of endoluminal scenes for controlling flexible robotic

endoscopic instruments,” *IEEE Transactions on Robotics*, vol. 42, pp. 1535–1555, 2026.

[2] X. Wang, B. Yang, M. Wei, Y. Chen, J. Zhang, and Y. Nie, “Endo-E2E-GS: End-to-end 3d reconstruction of endoscopic scenes using gaussian splatting,” *Displays*, vol. 92, p. 103353, 2026.

[3] C.-T. Hsieh, J. Shan, H. Fang, J. Hu, T. Ni, L. Han, Y. Ban, S. S. Cheng, and H. Wang, “ExtraGS: Enhancing endoscopic view extrapolation via diffusion-guided 3d gaussian splatting,” *arXiv preprint arXiv:2607.12785*, 2026.

[4] Y. He, P. Guo, M. Xu, Z. Li, A. Myronenko, D. Imans, B. Liu, D. Yang, M. Gu, Y. Ji, Y. Jin, R. Zhao, B. Shen, and D. Xu, “SurgWorld: Learning surgical robot policies from videos via world modeling,” *arXiv preprint arXiv:2512.23162*, 2025. [Online]. Available: <https://arxiv.org/abs/2512.23162v1>

[5] M. Allan, J. Mcleod, C. Wang, J. C. Rosenthal, Z. Hu, N. Gard, P. Eisert, K. X. Fu, T. Zeffiro, W. Xia, Z. Zhu, H. Luo, F. Jia, X. Zhang, X. Li, L. Sharan, T. Kurmann, S. Schmid, R. Sznitman, D. Psychogios, M. Azizian, D. Stoyanov, L. Maier-Hein, and S. Speidel, “Stereo correspondence and reconstruction of endoscopic data challenge,” *arXiv preprint arXiv:2101.01133*, 2021.

[6] S. Niklaus and F. Liu, “Softmax splatting for video frame interpolation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 5437–5446.

[7] Y. Zhao, C. Barnes, Y. Zhou, E. Shechtman, S. Amirghodsi, and C. C. Fowlkes, “GeoFill: Reference-based image inpainting with better geometric understanding,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2023, pp. 1776–1786.

[8] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, and V. Lempitsky, “Resolution-robust large mask inpainting with fourier convolutions,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022, pp. 2149–2159.

[9] X. Zou, Z. Zhang, D. Yu, W. Sun, W. Liu, D. Hang, W. Bao, and G. Zheng, “SSIFNet: Spatial-temporal stereo information fusion network for self-supervised surgical video inpainting,” *Computerized Medical Imaging and Graphics*, vol. 125, p. 102622, 2025.

[10] J. Guo, J. Wang, R. Wei, D. Kang, Q. Dou, and Y.-H. Liu, “UC-NeRF: Uncertainty-aware conditional neural radiance fields from endoscopic sparse views,” *IEEE Transactions on Medical Imaging*, vol. 44, no. 3, pp. 1284–1296, 2025.

[11] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. S. Huang, “Free-form image inpainting with gated convolution,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 4471–4480.

[12] T. Chu, J. Chen, J. Sun, S. Lian, Z. Wang, Z. Zuo, L. Zhao, W. Xing, and D. Lu, “Rethinking fast fourier convolution in image inpainting,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 23 195–23 205.

[13] W. Li, Z. Lin, K. Zhou, L. Qi, Y. Wang, and J. Jia, “MAT: Mask-aware transformer for large hole image inpainting,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 10 758–10 768.

[14] X. Ju, X. Liu, X. Wang, Y. Bian, Y. Shan, and Q. Xu, “BrushNet: A plug-and-play image inpainting model with decomposed dual-branch diffusion,” in *Computer Vision – ECCV 2024*. Springer Nature Switzerland, 2024, pp. 150–168.

- [15] L. Xie, D. Pakhomov, Z. Wang, Z. Wu, Z. Chen, Y. Zhou, H. Zheng, Z. Zhang, Z. Lin, J. Zhou, and C. Dong, "TurboFill: Adapting few-step text-to-image model for fast image inpainting," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 7613–7622.
- [16] F. X. Zhang, S. Chen, X. Xie, and H. P. H. Shum, "Depth-aware endoscopic video inpainting," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*, ser. Lecture Notes in Computer Science, vol. 15006. Springer Nature Switzerland, 2024, pp. 143–153.
- [17] O. Wiles, G. Gkioxari, R. Szeliski, and J. Johnson, "SynSin: End-to-end view synthesis from a single image," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 7467–7477.
- [18] Y. Chen, H. Xu, C. Zheng, B. Zhuang, M. Pollefeys, A. Geiger, T.-J. Cham, and J. Cai, "MVSplat: Efficient 3d gaussian splatting from sparse multi-view images," in *Computer Vision – ECCV 2024*, ser. Lecture Notes in Computer Science, vol. 15079. Springer Nature Switzerland, 2024, pp. 370–386.
- [19] H. Xu, S. Peng, F. Wang, H. Blum, D. Barath, A. Geiger, and M. Pollefeys, "DepthSplat: Connecting gaussian splatting and depth," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 16453–16463.
- [20] C. Cao, Y. Cai, Q. Dong, Y. Wang, and Y. Fu, "LeftRefill: Filling right canvas based on left reference through generalized text-to-image diffusion model," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 7705–7715.
- [21] N. Müller, K. Schwarz, B. Rössle, L. Porzi, S. R. Bulò, M. Nießner, and P. Kotschieder, "MultiDiff: Consistent novel view synthesis from a single image," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 10258–10268.
- [22] W. Tang, T. Chen, Y. Espinel, S. Farid, E. Buc, A. Bartoli, and S. Ali, "Point-guided latent diffusion model for novel view synthesis in laparoscopic liver surgery," *Healthcare Technology Letters*, vol. 12, no. 1, p. e70032, 2025.
- [23] Y. Wang, Y. Long, S. H. Fan, and Q. Dou, "Neural rendering for stereo 3d reconstruction of deformable tissues in robotic surgery," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*. Springer, 2022, pp. 431–441.
- [24] R. Zha, X. Cheng, H. Li, M. Harandi, and Z. Ge, "EndoSurf: Neural surface reconstruction of deformable tissues with stereo endoscope videos," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*. Springer, 2023, pp. 13–23.
- [25] C. Li, B. Y. Feng, Y. Liu, H. Liu, C. Wang, W. Yu, and Y. Yuan, "EndoSparse: Real-time sparse view synthesis of endoscopic scenes using gaussian splatting," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*, ser. Lecture Notes in Computer Science, vol. 15006. Springer Nature Switzerland, 2024, pp. 252–262.
- [26] H. Gou, C. Wang, J. Yang, Y. Liu, F. Jia, D. Xiao, F. Qin, and H. Luo, "Endo-GSMT: Endoscopic monocular scene reconstruction with dynamic gaussian splatting and motion tracking," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, ser. Lecture Notes in Computer Science, vol. 15968. Springer Nature Switzerland, 2025, pp. 213–223.
- [27] M. Minai, Y. Monno, M. Okutomi, and S. Suzuki, "Gastroendoscopy view synthesis: A new real dataset and evaluation," *arXiv preprint arXiv:2606.25427*, 2026.
- [28] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*. Springer International Publishing, 2015, pp. 234–241.
- [29] S. Yang, Q. Li, D. Shen, B. Gong, Q. Dou, and Y. Jin, "Deform3DGS: Flexible Deformation for Fast Surgical Scene Reconstruction with Gaussian Splatting," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*. Springer Nature Switzerland, 2024, pp. 132–142.
- [30] Y. Huang, B. Cui, L. Bai, Z. Guo, M. Xu, M. Islam, and H. Ren, "Endo-4DGS: Endoscopic Monocular Scene Reconstruction with 4D Gaussian Splatting," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*. Springer Nature Switzerland, 2024, pp. 197–207.
- [31] J. Guo, J. Wang, D. Kang, W. Dong, W. Wang, and Y.-H. Liu, "Free-SurGS: SfM-Free 3D Gaussian Splatting for Surgical Scene Reconstruction," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*. Springer Nature Switzerland, 2024, pp. 350–360.
- [32] J. Chen, X. Zhang, M. I. Hoque, F. Vasconcelos, D. Stoyanov, D. S. Elson, and B. Huang, "SurgicalGS: Dynamic 3D Gaussian Splatting for Accurate Robotic-Assisted Surgical Scene Reconstruction," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*. Springer Nature Switzerland, 2025, pp. 572–582.
- [33] J. Kaleta, W. Smolak-Dyżewska, D. Malarz, D. Dall'Alba, P. Korzeniowski, and P. Spurek, "PR-ENDO: Physically Based Relightable Gaussian Splatting for Endoscopy," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, vol. LNCS 15969. Springer Nature Switzerland, 2025, pp. 391–401.
- [34] L. Zhu, Z. Wang, J. Cui, Z. Jin, G. Lin, and L. Yu, "EndoGS: Deformable Endoscopic Tissues Reconstruction with Gaussian Splatting," in *MICCAI 2024 Workshops*. Springer, 2024, pp. 135–145.
- [35] Y. Liu, C. Li, H. Liu, C. Yang, and Y. Yuan, "Foundation Model-Guided Gaussian Splatting for 4D Reconstruction of Deformable Tissues," *IEEE Transactions on Medical Imaging*, vol. 44, no. 6, pp. 2672–2682, 2025.
- [36] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 10 684–10 695.
- [37] S. Luo, Y. Tan, S. Patil, D. Gu, P. von Platen, A. Passos, L. Huang, J. Li, and H. Zhao, "LCM-LoRA: A universal stable-diffusion acceleration module," *arXiv preprint arXiv:2311.05556*, 2023. [Online]. Available: <https://arxiv.org/abs/2311.05556>
- [38] Z. Li, J. Zhang, Q. Lin, J. Xiong, Y. Long, X. Deng, Y. Zhang, X. Liu, M. Huang, Z. Xiao, D. Chen, J. He, J. Li, W. Li, C. Zhang, R. Quan, J. Lu, J. Huang, X. Yuan, X. Zheng, Y. Li, J. Zhang, C. Zhang, M. Chen, J. Liu, Z. Fang, W. Wang, J. Xue, Y. Tao, J. Zhu, K. Liu, S. Lin, Y. Sun, Y. Li, D. Wang, M. Chen, Z. Hu, X. Xiao, Y. Chen, Y. Liu, W. Liu, D. Wang, Y. Yang, J. Jiang, and Q. Lu, "Hunyuan-DiT: A powerful multi-resolution diffusion transformer with fine-grained Chinese understanding," 2024. [Online]. Available: <https://arxiv.org/abs/2405.08748>
- [39] P. Charbonnier, L. Blanc-Féraud, G. Aubert, and M. Barlaud, "Two deterministic half-quadratic regularization algorithms for computed imaging," in *Proceedings of the 1st International Conference on Image Processing*, vol. 2, 1994, pp. 168–172.
- [40] A. Kendall and Y. Gal, "What uncertainties do we need in bayesian deep learning for computer vision?" in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [41] The iMED Challenge Organizers, "iMED Challenge 2026," MICCAI 2026 EndoVis Challenge, 2026, accessed: 2026-09-10. [Online]. Available: <https://imed-challenge.github.io/>
- [42] M. Allan, J. Mcleod, C. Wang, J. C. Rosenthal, Z. Hu, N. Gard, P. Eisert, K. X. Fu, T. Zeffiro, W. Xia *et al.*, "Stereo correspondence and reconstruction of endoscopic data challenge," *arXiv preprint arXiv:2101.01133*, 2021.
- [43] Q. Huynh-Thu and M. Ghanbari, "Scope of validity of PSNR in image/video quality assessment," *Electronics Letters*, vol. 44, no. 13, pp. 800–801, 2008.
- [44] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.
- [45] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 586–595.